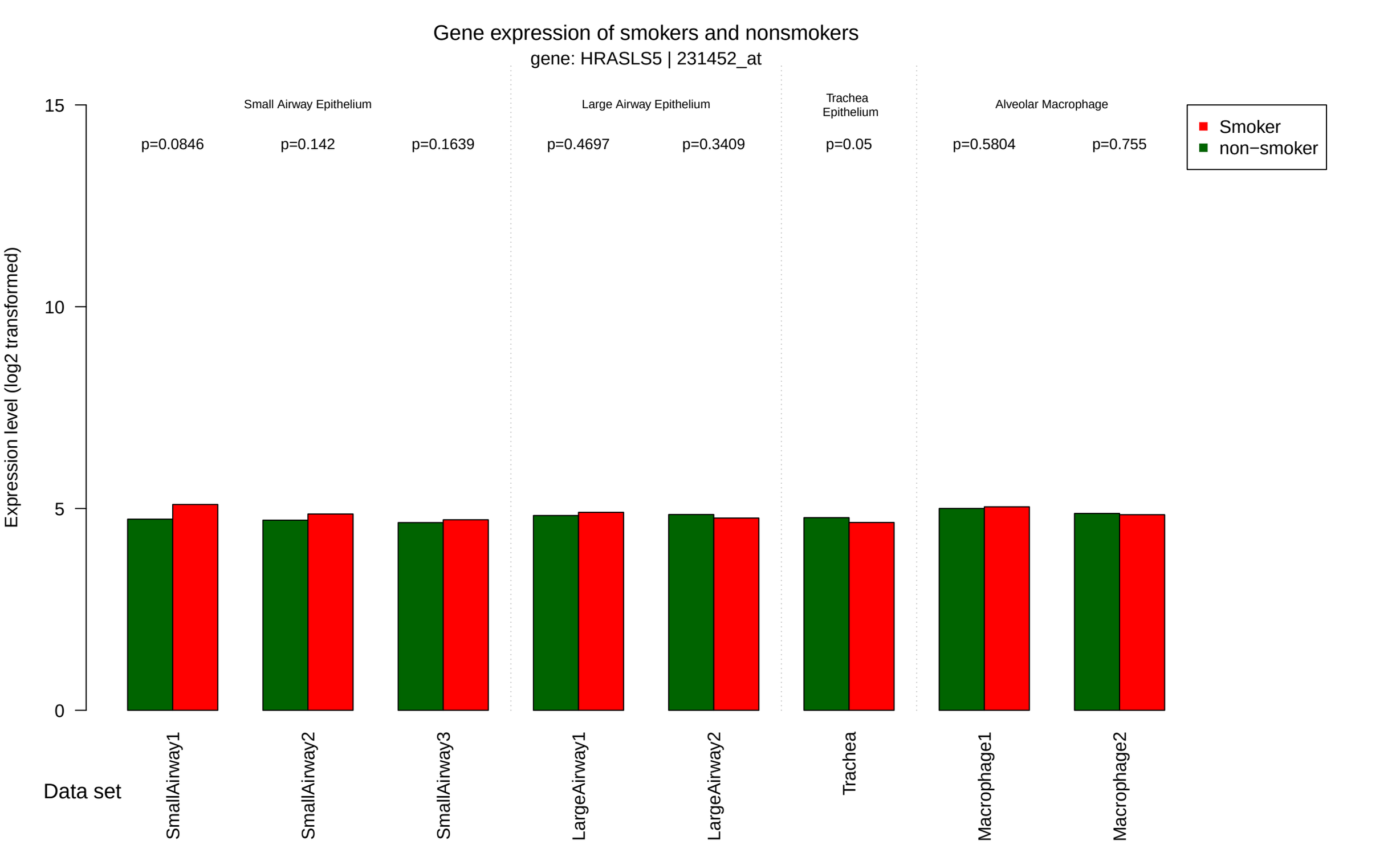
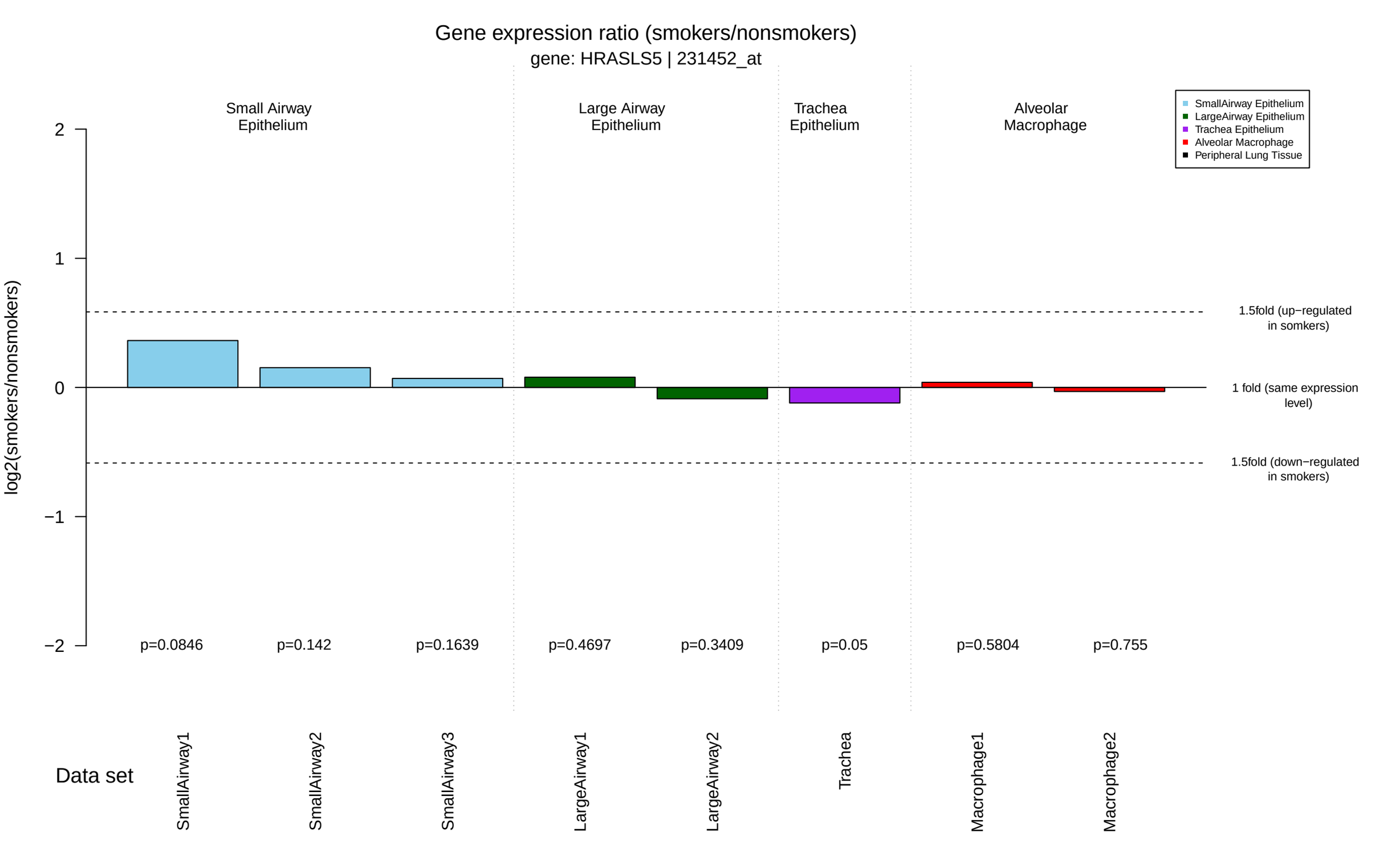
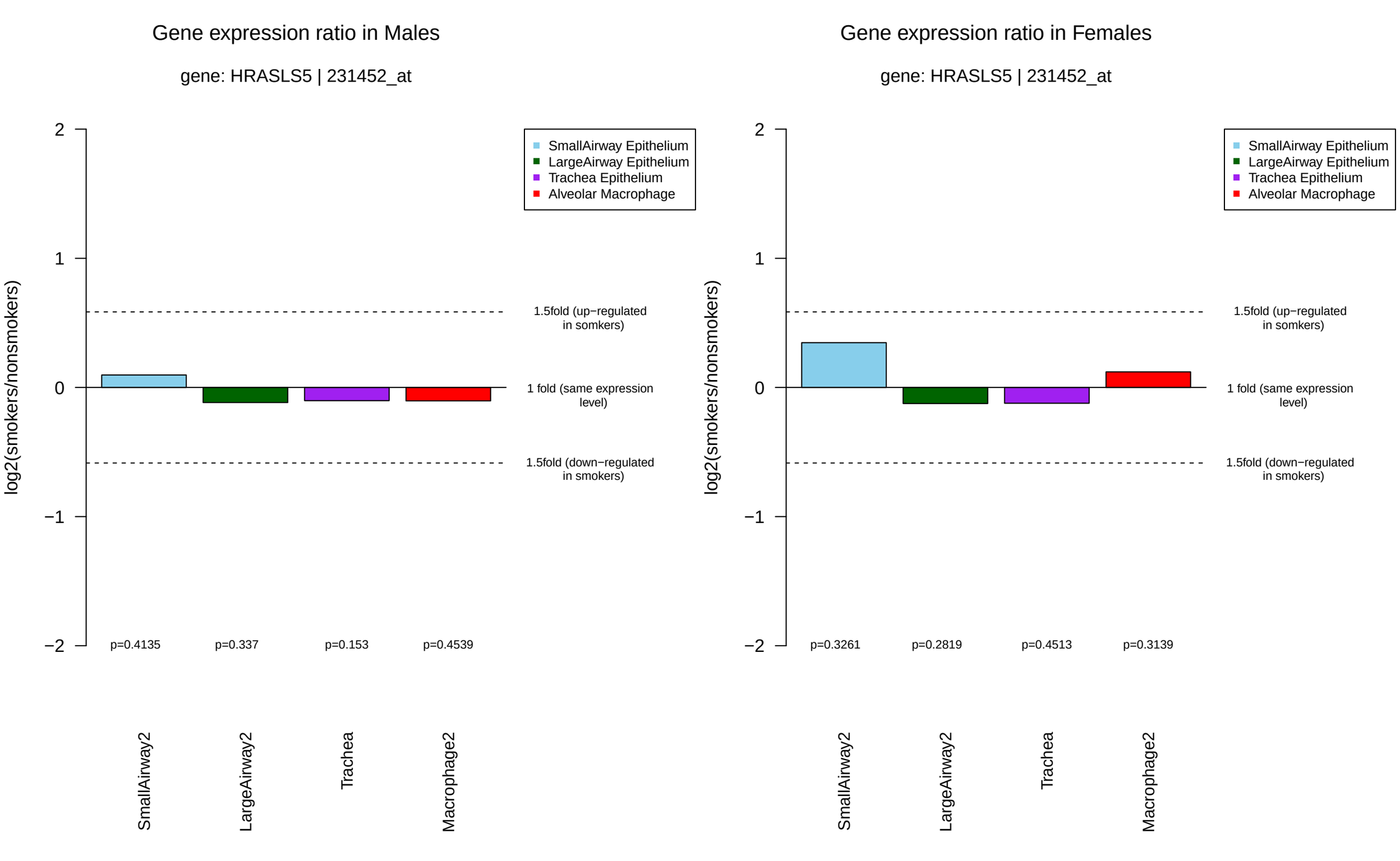
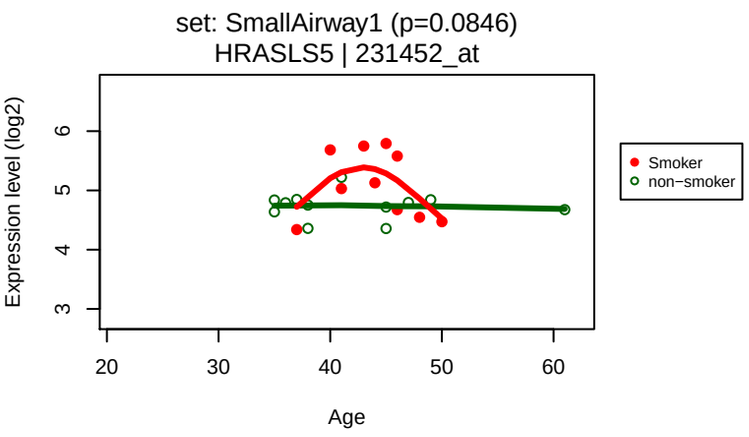
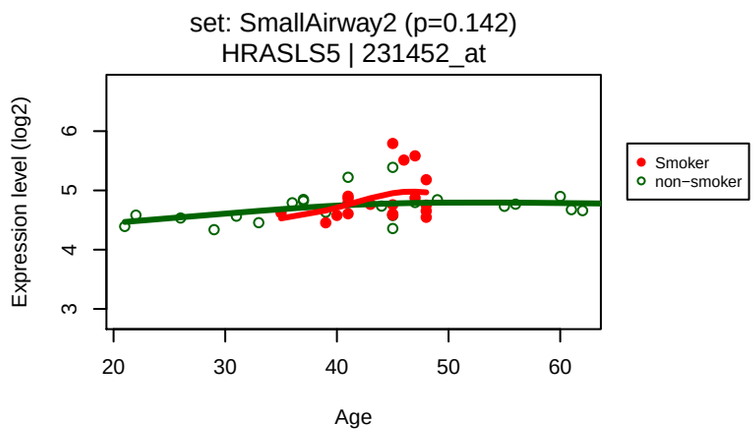
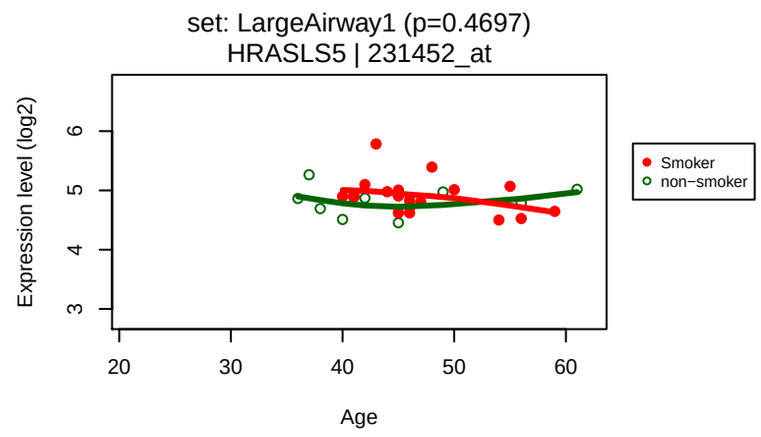

Gene expression of smokers and nonsmokers
gene: HRASLS5 | 231452_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0846
p=0.142
p=0.1639
p=0.4697
p=0.3409
p=0.05
p=0.5804
p=0.755
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

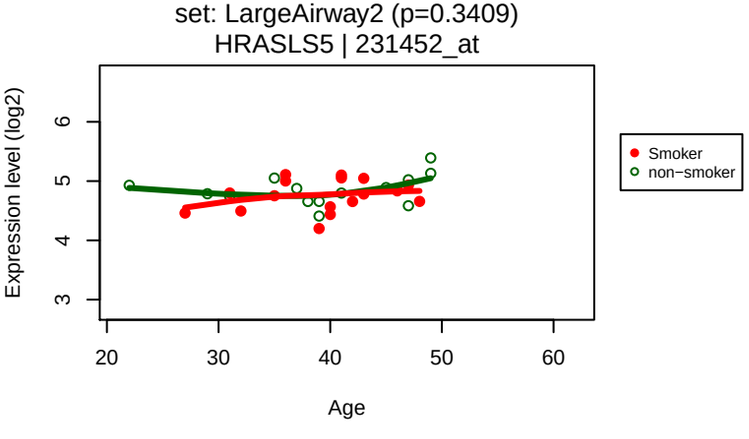

Gene expression ratio (smokers/nonsmokers)
gene: HRASLS5 | 231452_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0846
p=0.142
p=0.1639
p=0.4697
p=0.3409
p=0.05
p=0.5804
p=0.755
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

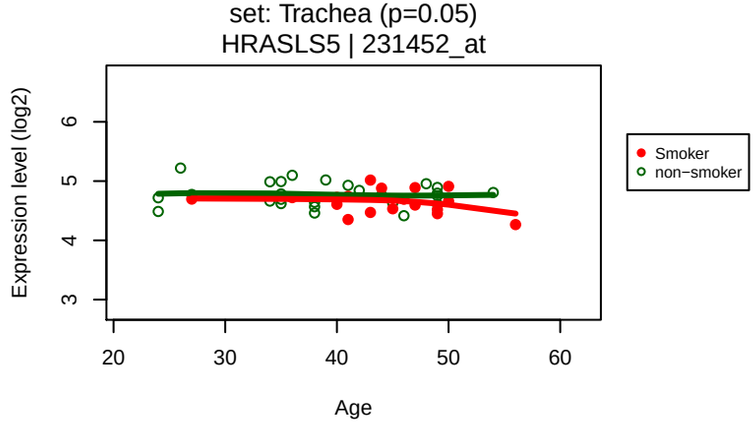

Gene expression ratio in Males
gene: HRASLS5 | 231452_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4135
p=0.337
p=0.153
p=0.4539
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HRASLS5 | 231452_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3261
p=0.2819
p=0.4513
p=0.3139
SmallAirway2
LargeAirway2
Trachea
Macrophage2

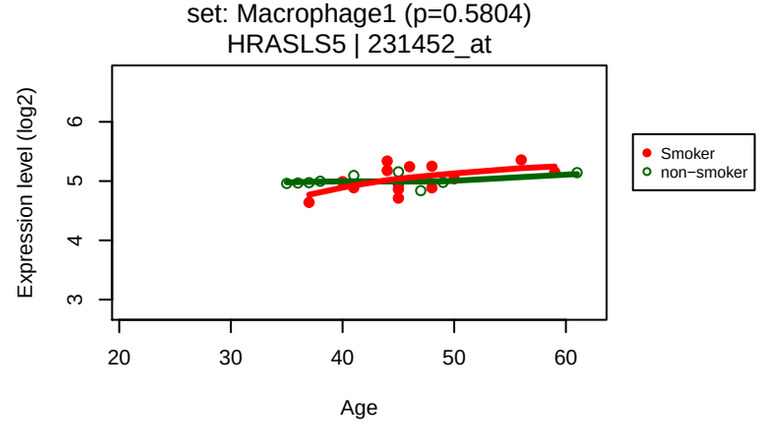

set: SmallAirway1 (p=0.0846)
HRASLS5 | 231452_at
Expression level (log2)
Age
Smoker
non−smoker

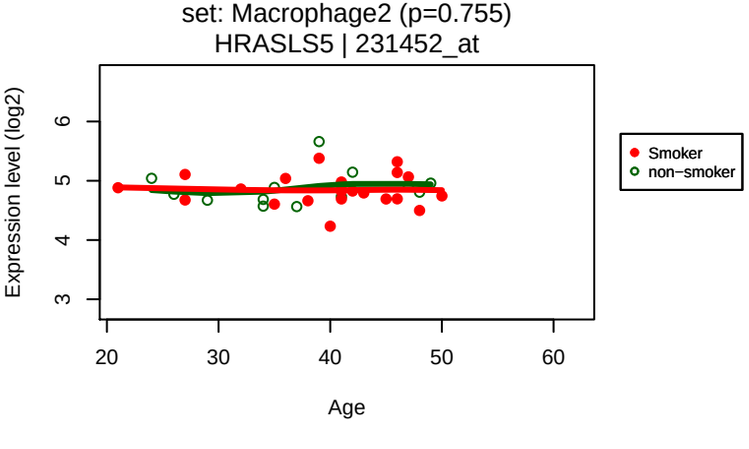

set: SmallAirway2 (p=0.142)
HRASLS5 | 231452_at
Expression level (log2)
Age
Smoker
non−smoker

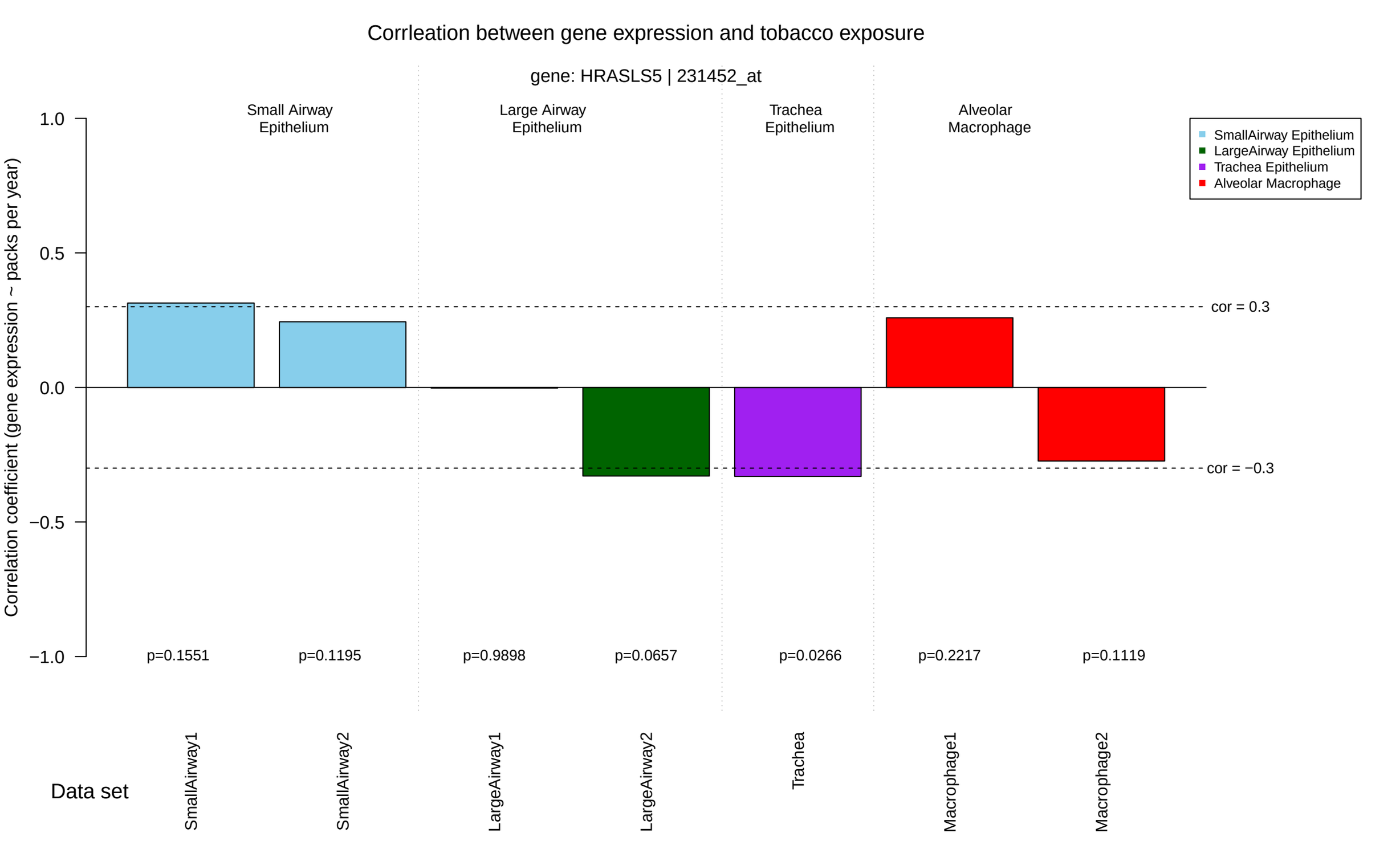

set: LargeAirway1 (p=0.4697)
HRASLS5 | 231452_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3409)
HRASLS5 | 231452_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.05)
HRASLS5 | 231452_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5804)
HRASLS5 | 231452_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.755)
HRASLS5 | 231452_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HRASLS5 | 231452_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1551
p=0.1195
p=0.9898
p=0.0657
p=0.0266
p=0.2217
p=0.1119
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2